W.S.61. **Acids and bases.** Name .....

Bases can neutralise (cancel out) acids. Bases that dissolve in water are called alkalis. A base reacts with an acid to form a salt and water :

ACID + BASE ⟹ SALT + WATER

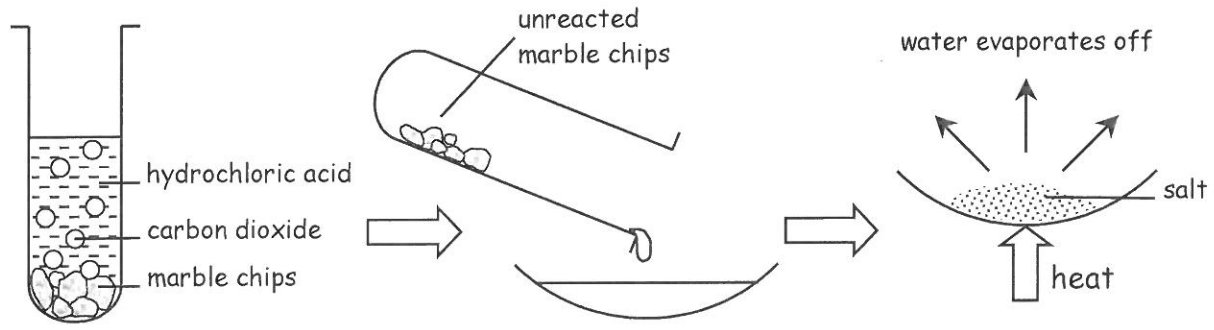
For example, if sodium hydroxide, which is a very strong alkali, is reacted with hydrochloric acid then sodium chloride (common salt) is formed.

Reaction with carbonates.

Carbonates are bases that contain the elements carbon and oxygen. They react with acids to form a salt, carbon dioxide gas and water. The reaction is fizzy due to the carbon dioxide gas given off :

ACID + CARBONATE ⟹ SALT + CARBON DIOXIDE + WATER

The experiment below shows the reaction between calcium carbonate (marble chips) and hydrochloric acid.



1) The marble chips react with the acid.

2) A solution of calcium chloride has formed.

3) **Calcium chloride** salt is left behind.

calcium carbonate + hydrochloric acid ⟶ calcium chloride + carbon dioxide + water

Exercise - Complete the missing words in the sentences and equations below.

1) A B _ _ _ is a chemical that can neutralise an acid.

2) Bases that dissolve in water are called A _ _ _ _ _ _

3) ACID + BASE ⟹ _____ + WATER

4) Sodium C _ _ _ _ _ _ _ is common salt.

5) Carbonates contain the elements carbon and O _ _ _ _ _

6) Carbonates react with acids to produce C _ _ _ _ _ D _ _ _ _ _ _

7) Marble chips will F _ _ _ in acid until it has all been neutralised.

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